O ‘ QUVQO ‘ LLANMA · 2 0 2 6 - YILKUZ

Chiziqli algebra I

1–4 haftalar

Har bir kurs fayli va tayinlangan darslik bo‘limlaridagi barcha savollar va formulalar — vektorlar, chiziqli kombinatsiyalar va span; skalyar va vektor ko‘paytmalar; matritsalar, matritsa algebrasi, determinantlar; rang va teskari matritsa — har bir masala to‘liq bosqichma-bosqich yechim bilan. Mustaqil o‘rganish va Test 1 ga tayyorgarlik uchun tuzilgan

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Manbalar: kurs konspektlari, darslar, ISA, TA sessiyalari; $G.$ Strang, Introduction to Linear Algebra $(4e)$ va Linear Algebra and its Applications $(4e)$

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1 Kurs xaritasi va hujjatdan qanday foydalanish

Hujjat bu 2026-yil kuz semestrining birinchi choragi bo‘lgan ”Chiziqli algebra I” kursining birinchi to‘rt haftasiga mo‘ljallangan to‘liq va mustaqbil qo‘llanmadir. U kurs repositoriyasining course-files/ papkasidagi har bir tegishli manbadan qo‘lda tuzilgan: haftalik konpektlar (Notes), ma’ruzalar prezentatsiyalari, amaliy mashg‘ulotlar varaqalari, mustaq ish uch vazifalar $(ISA), TA$ sessiyalarining masalalari va G. Strang darsliklarining belgilangan bo‘limlari. Masalalar so‘zma-so‘z takrorlangan; yechimlar to‘liq, bosqichma-bosqich, tekshirishlar bilan.

Hafta	Chiziqli algebra I
1	Vektorlar; chiziqli kombinatsiyalar; span; chiziqli mustaqillik
2	Skalyar ko‘paytma, burchaklar, proyeksiyalar; vektor ko‘paytma, yuzalar; matritsalar: amallar
3	Matritsa algebrasi; transponir; matritsalarni ko‘paytirish; determinantlar; aralash ko‘paytma
4	Matritsa rangi; Gauss-Jordan usulida teskari matritsa

Yozuv. Matritsat satr bo‘yicha yoziladi: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ — birinchi satr ab, ikkinchi cd; ustun-vektor $\begin{bmatrix} x \\ y \end{bmatrix}$, satr-vektor $\begin{bmatrix} x & y \end{bmatrix}$ ko‘rinishida yoziladi. Darajalar va indekslar ^{} va_ orqali yoziladi: A^{-1}, A^T, a_{11}.

2 Hafta 1 — Vektorlar (Chiziqli algebra I)

2.1 Chiziqli algebra I: 1-hafta formulalari katalogi

Vektorlar R^{n} dagi ustunlardir; 1-haftaning hammasi qo’shish, skalyarga ko’paytirish, birikmalar, span va mustaqillikdan quriladi.

$$(u+v)_i = u_i + v_i, \quad (c\,u)_i = c\,u_i \text{ (komponentalar bo’yicha amallar)} \tag{24}$$

$$c_1v_1 + \cdots + c_nv_n = \text{chiziqli birikma} \tag{25}$$

$$\text{span}\{v_1, \ldots, v_n\} = \text{bunday barcha birikmalar to’plami} \tag{26}$$

$$v_1, \ldots, v_n \text{ mustaqil} \Leftrightarrow c_1v_1 + \cdots + c_nv_n = 0 \Rightarrow c_1 = \cdots = c_n = 0 \tag{27}$$

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}, \quad \hat{x} = \frac{x}{\|x\|} \text{ (birlik vektor)} \tag{28}$$

$$u \text{ yo‌nalishda } L \text{ uzunlikdagi } v: \; v = \frac{L\,u}{\|u\|} \tag{29}$$

$$d(P,Q) = \|Q - P\| \tag{30}$$

2.2 Chiziqli algebra I, 1-hafta: ISA mashqlari — savollar va to‘liq yechimlar

$$[][] - 2 - 1$$

Butun vektorlar bo‘limida vektorlar ustunlardir; A qismida $Au =, v =$. 3 5

$$[][] - 2 - 1$$

1. $u = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$ **bo‘lsin.** $u + v, u - v$**, 2u, −3v, 2u − 3v ni hisoblang.**

[Manba: *ISALA*, hafta 1, mashq 1]

Yechim. (24) bo‘yicha komponent bo‘yicha: $u+v = \begin{bmatrix} -3 \\ 8 \end{bmatrix}$; $u-v = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$; 2u $= \begin{bmatrix} -4 \\ 6 \end{bmatrix}$; $-$3v $= \begin{bmatrix} 3 \\ -15 \end{bmatrix}$; 2u $-$ 3v $= \begin{bmatrix} -4 & + & 3 \\ 6 & + & 15 \end{bmatrix} = \begin{bmatrix} -1 \\ 21 \end{bmatrix}$.

2. Berilgan $a = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$**; toping** $a + b, a - b$**,** 3a + 2b**.**

$$\begin{bmatrix} 5 & -1 \\ 6 & 12 \end{bmatrix}$$

[Manba: *ISALA*, hafta 1, mashq 2]

Yechim. $a+b = \begin{bmatrix} -1 & + & 2 \\ 4 & & 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$; $a-b = \begin{bmatrix} -1 & & 2 \\ 4 & + & 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$; 3a + 2b $= [3\cdot(-1)+2\cdot 2;$ $3 \cdot 4 + 2 \cdot (-2)] = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$.

$$\begin{bmatrix} 2 & 6 & 12 \\ -4 & 8 & \end{bmatrix} \begin{bmatrix} 2 \\ 7 \end{bmatrix}$$

3. Toping x**, agar** $x + \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ -2 \end{bmatrix}$**.**

[Manba: *ISALA*, hafta 1, mashq 3]

Yechim. Ma’lum vektorni ikkala tomondan ayiramiz: $x = \begin{bmatrix} 7 & & 2 \\ 3 & + & 4 \\ -2 & & 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ -3 \end{bmatrix}$.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} - \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \\ -2 \end{bmatrix}$$

4. Toping x**, agar 3x − 2u** $= v$**, bunda** $u = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$**.**

[Manba: $ISALA$, hafta 1, mashq 4]

Yechim. Ajratamiz: 3x = v+ 2u = $\begin{bmatrix} 8 \\ 2 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$, shundan $x = [\frac{4}{3}; 4/3]$. Skalyarga ko'paytirish qo'shishga nisbatan taqsimlanuvchi, shuning uchun vektorni 3 ga bo'lish degani, uning har bir komponentini 3 ga bo'lishdir.

5. Vektorlarning Evklid uzunligini hisoblang: $u = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$, $w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

[Manba: $ISALA$, hafta 1, mashq 5]

Yechim. (28) ga ko'ra: $\|u\| = \sqrt{9} + 16 = 5$; $\|v\| = \sqrt{4} + 9 + 36 = 7$; $\|w\| = 0$ (faqat nol vektor nol uzunlikka ega).

6. $v = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$ **vektoriga bir xil yo'nalishda birlik vektorni toping va uning uzunligi 1 ga teng ekanini tekshiring.**

$$[-\frac{4}{5}; \frac{3}{5}]$$

[Manba: $ISALA$, hafta 1, mashq 6]

Yechim. $\|v\| = \sqrt{16} + 9 = 5$, shuning uchun $\hat{v} = v/5 = [-\frac{4}{5}; 3/5]$. Tekshirish: $\|\hat{v}\| = \sqrt{\frac{16}{25} + \frac{9}{25}} = 1$.

7. $v = \begin{bmatrix} 3 \\ -8 \end{bmatrix}$ **vektoriga bir xil yo'nalishda uzunligi 10 bo'lgan vektorni toping.**

[Manba: $ISALA$, hafta 1, mashq 7]

Yechim. Bu yerda $\|v\| = 5$, shuning uchun $\frac{10}{5} = 2$ miqdorida masshtablaymiz: javob 2v =

$$\begin{bmatrix} 6 \\ -16 \end{bmatrix}$$

$\|2v\| = 2 \cdot 5 = 10$ norma bir xilligi bo'yicha.

8. Vektorlarni ayirish va uzunlikdan foydalanib, $P = (1, -2, 3) va Q = (5, 1, -3)$ **orasidagi masofani toping.**

[Manba: $ISALA$, hafta 1, mashq 8]

Yechim. $PQ = Q - P = \begin{bmatrix} 5 \\ 1 \\ -3 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ -6 \end{bmatrix}$, shuning uchun $d(P, Q) = \|PQ\| = \sqrt{16} + 9 + 36 = \sqrt{61}$.

$$[5 - 1; 1 - (-2); -3 - 3]$$

$$\begin{bmatrix} 4 \\ 3 \\ -6 \end{bmatrix}$$

. Shuning uchun $d(P,\,Q) = \|PQ\| = \sqrt{4^2} + 3^2 + (-6)^2 = \sqrt{16} + 9 + 36 = \sqrt{61}$.

9. $v_1 =, v_2 =$ **bo'lsin.** $2v_1 + v_2,\ -v_1 + 3v_2,\ 4v_1 - 2v_2,\ 0v_1 + 5v_2$ **ni hisoblang. 2 3**

$$[][][][][]1 - 1 - 1 - 24 + 26$$

[Manba: $ISALA$, hafta 1, mashq 9]

Yechim. $2v_1 + v_2 =$

$$\begin{bmatrix} 2 & 2 \\ 6 & 3 \end{bmatrix}$$

; $-v_1 + 3v_2 =$

$$\begin{bmatrix} -1 & 3 \\ -3 & 9 \end{bmatrix}$$

; $4v_1 - 2v_2 =$

$$\begin{bmatrix} 4 & -2 \\ 12 & -6 \end{bmatrix}$$

; $0v_1 + 5v_2 =$ (nol skalyar v_1 ni to'liq yo'q qiladi).

$$\begin{bmatrix} 0 & 5 \\ 0 & 15 \end{bmatrix}$$

$5v_2 =$

$$\begin{bmatrix} 1 & -1 & 7 \\ -1 & -2 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 6 \\ 0 & 0 \end{bmatrix}$$

.

10. c_1, c_2 **ni toping shundayki** $c_1 + c_2 =$**; almashtirish orqali tekshiring. 1 2 1**

[Manba: $ISALA$, hafta 1, mashq 10]

Yechim. Komponentli tenglamalar: $c_1 - c_2 = 7vac_1 + 2c_2 = 1$. Birinchisini ikkinchisidan ayiramiz:

$$c_1 + c_2 = 1; c_1 - c_2 = 7; 2c_1 + c_2 = 1.$$

$3c_2 = -6$, demak $c_2 = -2vac_1 = 5$. Tekshirish:

$$\begin{bmatrix} 5 & & 2 \\ 5 & + & 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

.

11. $b = \begin{bmatrix}5\\1\\4\end{bmatrix}$ **vektorini** $b = c1 \cdot v1 + c2 \cdot v2 + c3 \cdot v3$ **ko'rinishida yozish mumkinmi, bunda** $v1 = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $v2 = \begin{bmatrix}0\\1\\1\end{bmatrix}$, $v3 = \begin{bmatrix}1\\1\\0\end{bmatrix}$**?**

[Manba: $ISALA$, hafta 1, mashq 11]

Yechim. Komponentlar $c_1 + c_3 = 5$, $c_2 + c_3 = 1$, $c_1 + c_2 = 4$ beradi. Birinchi ikkitasini qo'shib, uchinchisini ayiramiz: $2c_3 = 2$, demak $c_3 = 1, c_1 = 4, c_2 = 0$. Kombinatsiya mavjud:$b = 4v_1 + 0 \cdot v_2 + 1 \cdot v_3$. Yechimning noyobligi shuni ham ko'rsatadiki, uchta vektor chiziqli bog'liq emas.

12. $c \cdot \begin{bmatrix}-2\\3\end{bmatrix}$, $c \in R$ **bo'lgan barcha vektorlarni geometrik tasvirlang;** $\begin{bmatrix}4\\-6\end{bmatrix}$, $\begin{bmatrix}-2\\3\end{bmatrix}$, $\begin{bmatrix}6\\-8\end{bmatrix}$ **vektorlari bu** span **ga tegishlimi?**

[Manba: $ISALA$, hafta 1, mashq 12]

Yechim. Span — boshlang'ich nuqtadan o'tuvchi (−2, 3) yo'nalishidagi butun to'g'ri chiziq. Tekshirish: $\begin{bmatrix}4\\-6\end{bmatrix}$ — ha $(c = -2 : -2 \cdot (-2) = 4va - 2 \cdot 3 = -6)$; $\begin{bmatrix}-2\\3\end{bmatrix}$ — ha $(c = 1)$; $\begin{bmatrix}6\\-8\end{bmatrix}$ uchun −2c = 6 va 3c = −8, ya'ni $c = -3vac = -\frac{8}{3}$ — ziddiyat, demak u to'g'ri chiziqda yotmaydi.

13. $v_1 =, v_2 =$ **vektorlari chiziqli mustaqilmi?** $c_1v_1 + 26c_2v_2 = 0$ **bo'lgan nol bo'lmagan** c_1, c_2 **ni toping va geometrik tushuntiring.**

[Manba: $ISALA$, hafta 1, mashq 13]

Yechim. $v_2 = 3v_1$, shuning uchun $3v_1 - v_2 = 0$ — no trivial bog'liqlik $(c_1 = 3, c_2 = -1)$. Vektorlar bog'liq. Geometrik jihatdan ikkalasi ham boshlang'ich nuqtadan o'tuvchi bir to'g'ri chiziqda yotadi, shuning uchun ular birga hali ham faqat shu to'g'ri chiziqni hosil qiladi — bog'liqlik bitta vektor yangi yo'nalish bermasligini anglatadi.

$$\begin{bmatrix}1 & 0\\1 & \end{bmatrix}$$

14. $v1 = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $v2 = \begin{bmatrix}0\\1\\1\end{bmatrix}$, $v3 = \begin{bmatrix}1\\1\\2\end{bmatrix}$ **vektorlari chiziqli mustaqilmi? Agar bog'liq bo'lsa, birini boshqalari orqali ifodalang.**

[Manba: $ISALA$, hafta 1, mashq 14]

Yechim. $v_3 = \alpha v_1 + \beta v_2$ deb faraz qilamiz: komponentlar $\alpha = 1$ (birinchi), $\beta = 1$(ikkinchi), $\alpha+\beta = 2$(uchinchi) — mos keladi. Demakv$_3 = v_1+v_2$, no trivial bog'liqlik mavjud va to'plam bog'liq. Faqat ikkita mustaqil yo'nal mavjud; span — tekislik.

$$\begin{bmatrix}1\\1\\1\end{bmatrix}$$

15. $v_1 =, v_2 =$ **bo'lsin. (1) Ular** R^2 **ni hosil qiladimi?** $(2)b =$ **uchun**

$$21[]b_27$$

c_1, c_2 **ni toping shundayki** $c_1v_1 + c_2v_2 = b$**. (3) formuladan ifodalang. 4**
[Manba: $ISALA$, hafta 1, mashq 15]

Yechim. $(1) - (2)c_1 + 3c_2 = b_1,\ 2c_1 + c_2 = b_2$ tenglamalarini yechamiz. Chiqarib: $c_1 = b_1 - 3c_2, u$ holda $2b_1 - 6c_2 + c_2 = b_2$, demak $c_2 = (2b_1 - b_2)/5vac_1 = (-b_1 + 3b_2)/5$. Formulalar har qanday (b_1, b_2) uchun aniqlangan, shuning uchun ha, ikkita vektor R^2 ni hosil qiladi. $(3)c_1 = (-7 + 12)/5 = 1$, $c_2 = (14 - 4)/5 = 2$: $b = v_1 + 2v_2$.

$$Q = uv = (2 : a = V) + 2Vo.$$

16. $u_1 =, u_2 =$ **bo'lsin. (1) Ular** R^2 **ni hosil qiladimi? (2) 2 3**

$$(1)u_2 = 6u_1, po$$

span**{**u_1, u_2 **} ni geometrik ravishda tasvirlang; (3) unga tegishlimi? −4 2**
[Manba: $ISALA$, hafta 1, mashq 16]

Yechim. Nisbatlarni solishtirish $-6/(-1) = 6va\frac{3}{2}$ ko'rsatadiki, bosilgan juftlik proporsional emas, shuning uchun bosilgandek uR^2 ni qamrab oladi; mashqning standart versiyasida $u2 = 3u1 = \begin{bmatrix}-3\\6\end{bmatrix}$ — uni yechamiz. $(1)u2 = 3u1$, shuning uchun span — faqat span$\{u1\}$. (2) Geometrik jihatdan bu (−1, 2) yo'nalishidagi boshidan o'tgan yagona to'g'ri chiziq; ikki kollinear vektor hech qachon tekislikni qamrab olmaydi. $(3)\begin{bmatrix}8\\-4\end{bmatrix}$ uchun: $-c = 8$ va 2c $= -4$, ya'ni $c = -8vac = -2$ — ziddiyat, demak yo'q; $\begin{bmatrix}1\\2\end{bmatrix}$ uchun: $-c = 1$ va 2c $= 2$, ya'ni $c = -1vac = 1$ — ziddiyat, demak yo'q.

$$y = (\frac{1}{2})\begin{bmatrix}1 & 0\\1 & 1\end{bmatrix}x + \begin{bmatrix}1\\0\end{bmatrix}(16)$$

17. Faraz qilaylik $v1 = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $v2 = \begin{bmatrix}0\\1\\1\end{bmatrix}$, $v3 = \begin{bmatrix}1\\1\\0\end{bmatrix}$**.** $(1)c1v1 + c2v2 + c3v3 = b$ **ni ixtiyoriy** $b = \begin{bmatrix}b1\\b2\\b3\end{bmatrix}$ **uchun yeching; (2) nima uchun bu uch vektor** R^3 **ni qamrab olishini**

tushuntiring; $(3) b = \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$ **ni ifodalang.**

$$c_1 = (b_1 + b_2 - b_3)/2$$

ixtiyoriyb $= b_2$ **uchun yeching; (2) uchta vektor** R^3 **ni egallaydigan sababini tushuntiring;**

$$c_1 = (b_1 + b_2 - b_3)/2$$

[Manba: $ISALA$, hafta 1, mashq 17]

Yechim. (1) Komponentlar: $c_1+c_3 = b_1$, $c_2+c_3 = b_2$, $c_1+c_2 = b_3$. Uchtasini qo'shamiz: $2(c_1 + c_2 + c_3) = b_1 + b_2 + b_3$, demak $c_1 + c_2 + c_3 = {}^{b1+b2+b3}$. Keyin $c_1 = {}^{b1-b2+b3}$, $22c_2 = {}^{-b1+b2+b3}$, $c_3 = {}^{b1+b2-b3}$. (2) Yechim istalgan o'ng tomon uchun mavjud va yagona, shuning uchun span — butun R^3 (vektorlar mustaqil). $(3) c_1 = {}^{4-2+6} = 24$, $c_2 = {}^{-4+2+6} = 2$, $c_3 = {}^{4+2-6} = 0$:$b = 4v_1 + 2v_2$. 2 2

$$b = 4v_1 + 2v_2$$

18. Detalga uchta kuch ta'sir qiladi: $F_1 = N, F_2 = N, F_3 =$

$$\begin{bmatrix} 10 \\ 25 \\ -5 \end{bmatrix}$$

N**. (1) Rivojlantiruvchini toping; (2) detali muvozanatga keltiruvchi**F_4 **ni toping:**$F_1 + F_2 + F_3 + F_4 = 0$**.**

$$\begin{bmatrix} 40 \\ -15 \\ -20 \end{bmatrix}$$

[Manba: $ISALA$, hafta 1, mashq 18]

Yechim. (1) Komponent bo'yicha qo'shamiz: $F_1 + F_2 + F_3 =$

$$\begin{bmatrix} 10 \\ 25 \\ -5 \end{bmatrix}$$

$= 50N$. (2) Muvozanat uchun $F_4 = -(F_1 + F_2 + F_3) =$

$$\begin{bmatrix} -40 \\ 15 \\ 20 \end{bmatrix}$$

N kerak: bu rivojlantiruvchi vektorga aynan qarama-qarshi, shunda umumiy kuch nolga aylanadi.

2.3 Chiziqli algebra I, hafta 1: TA sessiya masalalari (alternativ to'plam) — savollar va to'liq yechimlar

TA to'plami sakkizta qisqa masaladan iborat, har biri rasmiy yechimga ega; ikkalasini ham keltiramiz, taxminlarni batafsil ochib berib.

1. $a = (-1, 3, 2), b = (4, 0, -2)$**. Hisoblanga** $+b$ **va 3a** $-b$**;** a **vab** orasidagi o'rtacheni toping.**

[Manba: Zadachi TA, nedelya 1, zadacha 1]

Yechim. $a + b = (3, 3, 0)$; 3a $-b = (-3 - 4,\ 9 - 0,\ 6 + 2) = (-7,\ 9,\ 8)$. O'rta — o'rtacha $^{a+b} = (\frac{3}{2}, 3_2, 0)$ — bu kesma o'rtasining skalyar formuladagi o'rtacha olish g'oyasi bilan bir xil.

2. Berilganp $+q = (5,\ 7)vap - q = (-1,\ 3)$**;**$pvaq$ **ni toping.**

[Manba: Zadachi TA, nedelya 1, zadacha 2]

Yechim. Ikkita tenglamani qo'shamiz: 2p $= (4, 10)$, demakp $= (2, 5)$. Ayiramiz: 2q $= (6, 4)$, demakq $= (3, 2)$. Tekshirish: $(2,\ 5) + (3,\ 2) = (5,\ 7)va(2,\ 5) - (3,\ 2) = (-1,\ 3)$. «Qo'shib ayirish» usuli — bu ikkita skalyar tenglamani bir vaqtda yechish vektor analogi.

3. au $= (1,\ 2,\ -1), v = (2,\ -1,\ 0)$**. Hisoblang** $-u+$ **3v; yozing cu** $+$ **dv komponent bo'yicha; cu** $+dv(5, 0, -1)$ **ga teng bo'lishi mumkinmi?**

[Manba: Zadachi TA, nedelya 1, zadacha 3]

Yechim. $-u+$ 3v $= (-1,\ -2,\ 1) + (6,\ -3,\ 0) = (5,\ -5,\ 1)$. Komponent bo'yicha cu $+dv = (c + 2d,$ 2c $-d, -c)$. (5, 0, −1) ga yetish uchun: uchinchi komponent $c = 1$ bo'lishini talab qiladi; keyin $c+$ 2d $= 5d = 2$ beradi; va haqiqatan ham 2c $-d = 0$. Javob: ha,$u+$ 2v $= (5, 0, -1)$.

4. ar $= (2,\ -1), s = (-6,\ 3)$**. Barcha kombinatsiyalarni cr** $+ds$ **ni tasviring; vektorlar** R^2 **ni to'ldiradimi? Bog'liqlik munosabatini toping.**

[Manba: TA masalalari, 1 hafta, 4-masala]

Yechim. $s = -3$r bo'lgani uchun, har qanday kombinatsiya $(c - 3d)r$ ko'rinishida bo'ladi — bu har doim r ning karrali. Span — bu orqali o'tgan to'g'ri chiziq: vektorlar tekislikni cho'zmaydi. 3r $+s = 0$ munosabati trivialemas, bu bog'liqlikni isbotlaydi (qarang (27)).

5. $v = (1, 0, 1), w = (1, 2, 0)$ **deb oling. cv** $+dw = (x, y, z)$ **yozing;** x, y, z **ni bog'laydigan bitta tenglamani toping;** $b = (4, 6, 1)$ **ni tekshiring.**

[Manba: TA masalalari, 1 hafta, 5-masala]

Yechim. cv $+dw = (c+d,$ 2d, $c)$. O'qib: $z = cvay =$ 2d, demak $x = z+\frac{y}{2}$ — bu tekislikni tasvirlaydigan yagona chiziqli tenglama. uchun: o'ng tomon $= 1+\frac{6}{2} = 4 = x$, demak b tekislikda joylashgan; $z = 1, y = 6$ dan: $c = 1, d = 3$.

6. c, d **ni shunday topingki,** $c(1,\ 2) + d(3,\ -1) = (7,\ 3)$**; komponent bo'yicha tenglamalarni yozing va yeching.**

[Manba: TA masalalari, 1 hafta, 6-masala]

Yechim. Komponentlar: $c+$ 3d $= 7$ va 2c $-d = 3$. Ikkinchidan $d =$ 2c $-$ 3; almashtiramiz: $c+$6c$-9 = 7$, demak 7c $= 16$, $c = \frac{16}{7}$, $d = 32_7 - 3 = 11_7$. Ikkala komponent ham aniq tekshiriladi.

7. $u = (1, 1, 0), v = (0, 1, 1), w = (1, 2, 1)$**.** w **yangi yo'nalish qo'shadimi?** span**{**u, v, w**}** **ni tasvirlang; nolga teng bo'lgan munosabatni keltiring.**

[Manba: TA masalalari, 1 hafta, 7-masala]

Yechim. $w = u + v$ (komponent bo'yicha tekshirish: $1+0=1, 1+1=2, 0+1=1$), shuning uchun w ortiqcha. $u + v - w = 0$ munosabati trivialemas, demak uchta vektor bog'liq va span — bu faqat $uvav$ (ular karrali emas) cho'zgan tekislik.

8. Ikkita vektor butun R^3 ni cho'zishi mumkinmi? Uchta vektor R^3 ni cho'zishi mumkin va bog'liq bo'lishi mumkinmi? "Yangi yo'nalishlar" orqali tushuntiring.

[Manba: TA masalalari, 1 hafta, 8-masala]

Yechim. Ikkita vektor maksimal tekislikni cho'zadi — har bir vektor maksimal bitta yangi yo'nalish beradi, ikki yo'nalish bilan uch o'lchamli fazoni to'ldirib bo'lmaydi. Uchta bog'liq vektor ham maksimal tekislikni cho'zadi: bog'liqlik bittasining qolganlarining kombinatsiyasi ekanligini va uchinchi yo'nalish qo'shmasligini anglatadi. R^3 ga erishish uchun uchta mustaqil yo'nalish kerak; shuning uchun uchta bog'liq vektor hech qachon R^3 ni cho'zmaydi.

3 2-Hafta — Skalyar va vektor ko'paytmalari, matritsalarga kirish (Lineal algebra I)

3.1 Lineal algebra I, 2-hafta: formulalar katalogi

2-hafta vektor geometriyasining ikki ko'paytmasi va birinchi matritsa tushunchalarini qo'shadi.

$$u \cdot v = u_1 v_1 + \cdots + u_n v_n = u^T v, \quad \cos\theta = \frac{u \cdot v}{\|u\|\,\|v\|} \tag{45}$$

$$u \perp v \Leftrightarrow u \cdot v = 0, \quad \text{proj}_v\, u = \frac{u \cdot v}{v \cdot v}\, v \tag{46}$$

$$u \times v = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}, \quad \|u \times v\| = \|u\|\,\|v\|\,|\sin\theta| \text{ (yuza)} \tag{47}$$

$$u \times v = -(v \times u), \quad (u \times v) \perp u, v \tag{48}$$

$$A = [a_{ij}] \in \mathbb{R}^{m \times n}, \quad (A+B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c\,a_{ij} \tag{49}$$

$$Ax = x_1 a_1 + \cdots + x_n a_n \text{ (ustunlar rasmi)} \tag{50}$$

bunda a_j — A ning j-ustuni. O'lchamlar mos bo'lishi shart: Ax faqat $Am \times nvax \in R^n$ bo'lganda mavjud, natija R^m da yotadi. Diagonal matritsa $D = diag(d_1, \ldots, d_n)har$ bir koordinatani masshtablaydi: $Dx = (d_1x_1, \ldots, d_nx_n)^T$.

3.2 Lineal algebra I, 2-hafta: ma’ruzadagi auditoriya masalalari — savollar va to’liq yechimlar

[3][]4

1. $u =, v =$**.** $u \cdot v$ **ni hisoblang; burchakni tasniflang.** -2 5

[Manba: LA ma’ruzasi, 2-hafta, 1-masala]

Yechim. $u \cdot v = 3 \cdot 4 + (-2) \cdot 5 = 12 - 10 = 2 > 0$. Musbat skalyar ko‘paytma $\cos\theta > 0$ degani, ya‘ni burchak o‘tkir.

2. $u = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}$ **bo‘lsin:** $\|u\| v a dist(u, v)$ **ni toping.**

[Manba: LA ma’ruzasi, 2-hafta, 2-masala]

Yechim. $\|u\| = \sqrt{4+1+4} = 3$. $u - v = \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix}$, demak $dist(u,\, v) = \sqrt{9+16+4} = \sqrt{29}$.

3. $u = \begin{bmatrix} 2 \\ k \\ -1 \end{bmatrix}$ **vektoriga perpendikular bo’ladigan** $v = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$ **uchun** k **ni toping.**

[Manba: LA ma’ruzasi, 2-hafta, 3-masala]

Yechim. Perpendikulyarlik — bu $u \cdot v = 0$: $6 + k - 4 = 0$, demak $k = -2$.

$$Pemenne.OpToroHasIbHOCT'—3TO0U \cdot V = 0 : 6 + k - 4 = 0, 3HauUTk = -2.$$

4. $u = vav =$ **o‘rtasidagi burchakni toping.** 1 -1

[Manba: Lektsiya LA, nedelya 2, zadacha 4]

Yechim. $u \cdot v = 1 - 1 = 0$: vektorlar perpendikulyar, $\theta = 90^\circ$ — formulalar kerak emas, skalyar ko‘paytma nolga teng bo‘lganda.

$$\cos\varphi = (u \cdot v)/(||u||||v||)$$

5. $u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ $niv = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ **ga proyeksiyaling.**

[Manba: Lektsiya LA, nedelya 2, zadacha 5]

Yechim. $u \cdot v = 6, v \cdot v = 2$, shuning uchun $proj_vu = (\frac{6}{2})v = 3\text{v} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$. Qoldiq $u-$ $3\text{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ shartni qanoatlantiradi: $(u - 3v) \cdot v = 0$ — parallel va perpendikular qismlarga ajralish.

6. $u = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$ **uchun** $u \times v$ **ni hisoblang va ortogonallikni tekshiring.**

$$2 \cdot 2 - 0 \cdot (-1)[4]$$

[Manba: Lektsiya LA, nedelya 2, zadacha 6]

Yechim. (47) ga ko‘ra:u ×v $= 0 \cdot 3 - 1 \cdot 2 = -2$. Tekshirish: $1 \cdot 4 + 2 \cdot (-2) + 0 \cdot$

$$u \times v = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \times \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} = [0 \cdot 0 - 2 \cdot 2;$$

$$2 \cdot (-1) - 2 \cdot 0; 2 \cdot 2 - 0 \cdot (-1)] = \begin{bmatrix} -4 \\ -2 \\ 4 \end{bmatrix} (16)$$

$(-7) = 0va3 \cdot 4 + (-1)(-2) + 2 \cdot (-7) = 0$ — ikkala kirishga ham ortogonal. 2 1

7. Tomonlari $a = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$ **bo‘lgan parallelogramm yuzini toping.**

$$0 \cdot (-1) - 1 \cdot 3 - 3$$

[Manba: Lektsiya LA, nedelya 2, zadacha 7]

Yechim. $a \times b = [0 \cdot (-1) - 1 \cdot 3;\ 1 \cdot 1 - 2 \cdot (-1);\ 2 \cdot 3 - 0 \cdot 1] = \begin{bmatrix} -3 \\ 3 \\ 6 \end{bmatrix}$, shuning uchun yuza

$\|a \times b\| = \sqrt{9} + 9 + 36 = \sqrt{54} = 3\sqrt{6}$.

$$2 \cdot 3 - 0 \cdot 1 = 6$$

8. $A =$ **uchun o‘lchamni ko‘rsating,** a_{13}, a_{22} **ni toping va ustunlarni yozing** $035a_1, a_3$**.**

[Manba: Lektsiya LA, nedelya 2, zadacha 8]

Yechim. A o‘lchami 2×3 (avval qatorlar). $a_{13} = 4(1-$qator, 3-ustun$), a_{22} = 3.a_1 = , a_3 =$. 0 5

$$A = \begin{bmatrix} 2 & 4 & \\ 0 & 3 & 5 \end{bmatrix}$$

9. $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 \\ -1 & 5 \end{bmatrix}$ **bo‘lsin. 2A** $-B$ **ni toping.**

[Manba: Lektsiya LA, nedelya 2, zadacha 9]

Yechim. 2A $= \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$; element bo‘yicha ayiramiz: 2A $-B = [2-4, -4-1;\ 6-(-1),$ $0-5] = \begin{bmatrix} -2 & -5 \\ 7 & -5 \end{bmatrix}$.

10. A **o‘lchami** $3\times 2, B$ **—** $2\times 3, x\in R^2, y\in R^3$**.** Ax, Ay, Bx, By **dan qaysilari aniqlangan?**

[Manba: Lektsiya LA, nedelya 2, zadacha 10]

Yechim. Ax: $(3\times 2)(2\times 1)OK$, natija R^3 da. By: $(2\times 3)(3\times 1)OK$, natija R^2 da. Ay mos kelmaydi $(2\neq 3)vaBx$ mos kelmaydi $(3\neq 2)$: ikkalasi ham aniqlanmagan. Faqat ichki o‘lchamlar mos kelishi kerak. 4

11. $D=diag(2,\,-1,\,3),\ x=\begin{bmatrix}4\\5\\-2\end{bmatrix}$ **bo’lsin.** $DxvaI$**3x ni toping.**

[Manba: Lektsiya LA, nedelya 2, zadacha 11]

Yechim. $Dx=[2\cdot 4;\,-1\cdot 5;\,3\cdot(-2)]=\begin{bmatrix}8\\-5\\-6\end{bmatrix}$: har bir koordinata o’z diagonal elementi bilan masshtablanadi. I3x $=x=\begin{bmatrix}4\\5\\-2\end{bmatrix}$: birlik matritsa hech narsa qilmaydi.

$$3\cdot(-2)=-6$$

. $I_3x=x=5$: birlik matritsa hech narsa qilmaydi. -2

12. $A=\begin{bmatrix}1&2&0\\-1&3&4\\4&0&8\end{bmatrix}, x=\begin{bmatrix}2\\-1\\0\end{bmatrix}$ **bo’lsin.** Ax **ni ustunlar tasviri va qatorlar skalyar ko’paytmalari orqali hisoblang.**

$$\begin{bmatrix}1&2&0\\-1&3&4\\0&0&8\end{bmatrix}$$

[Manba: Lektsiya LA, nedelya 2, zadacha 12]

Yechim. Ustunlar: $2\cdot\begin{bmatrix}1\\-1\\4\end{bmatrix}+(-1)\cdot\begin{bmatrix}2\\3\\0\end{bmatrix}+0\cdot\begin{bmatrix}0\\4\\8\end{bmatrix}=\begin{bmatrix}0\\-5\\8\end{bmatrix}$. Qatorlar: $\begin{bmatrix}1&2\end{bmatrix}\cdot x=0$, $\begin{bmatrix}-1&3\end{bmatrix}\cdot x=-5$, $\begin{bmatrix}4&0\end{bmatrix}\cdot x=8$; ustunma-ustun xuddi shu $\begin{bmatrix}0\\-5\\8\end{bmatrix}$. Ikki tasvir har doim mos keladi.

$$13.ChiziqliaylantirishL:R^3\to R^3L(x)Axformulasibilananiqlanadi,$$
$$bundaA=[-130;2-14;0$$

12].

13. $A = va Ax =$ **berilgan;** x **ni toping. 3 −1 5**

[Manba: Lektsiya LA, nedelya 2, zadacha 13]

Yechim. Komponentlar: $x_1 + 2x_2 = 7$, $3x_1 - x_2 = 5$. Ikkinchisidan $x_2 = 3x_1 - 5$;

$$x = [\frac{17}{7}]$$

almashtiramiz: $x_1 + 6x_1 - 10 = 7$, demak $7x_1 = 17$, $x_1 = \frac{17}{7}$, $x_2 = 16_7$. Shunday qilibx =. 16/7

14. Nol bo‘lmagan vektorlar $u+vvau-v$ **ortogonal.** $\|u\|va\|v\|$ **haqida nima deyish mumkin?**

[Manba: Lektsiya LA, nedelya 2, zadacha 14]

Yechim. Ortogonallik $0 = (u + v) \cdot (u - v) = \|u\|^2 - \|v\|^2$ (kesishuvchi hadlar bekor qilinadi), bundan $\|u\| = \|v\|$: teng uzunliklar, vau vav — to‘g‘ri burchakli parallelogrammning yarim diagonallari.

3.3 Chiziqli algebra I, nedelya 2: mashqlar ISA — savollar va to‘liq yechimlar

1. $u = (-3,\ 2)^T, v = (4,\ 5)^T$ **bo‘lsin.** $u \cdot v$ **ni hisoblang; burchak o‘tkir, to‘g‘ri yoki o‘tkirmi?**

[Manba: $LAISA$, nedelya 2, mshr. 1]

Yechim. $u \cdot v = -12 + 10 = -2 < 0$, demak $\cos\theta < 0$: burchak o‘tkir.

2. $\mathbf{aa} = (1,\ -2,\ 4)^T, b = (3,\ 0,\ -1)^T$**. Topinga** $\cdot b$, $\|a\|$, $\|b\|$, $\|a - b\|$**.**

[Manba: $ISALA$, hafta 2, mashq 2]

Yechim. $a \cdot b = 3 + 0 - 4 = -1$; $\|a\| = \sqrt{1 + 4 + 16} = \sqrt{21}$; $\|b\| = \sqrt{9 + 0 + 1} = \sqrt{10}$; $a - b = (-2,\ -2,\ 5)^T$, $\|a - b\| = \sqrt{4 + 4 + 25} = \sqrt{33}$.

3. $u = (2, k, 3)^T$ **vektori** $v = (-1, 4, 2)^T$ **vektoriga perpendikulyar bo’ladigan** k **ni toping.**

[Manba: $ISALA$, hafta 2, mashq 3]

Yechim. $u \cdot v = -2 + 4\mathrm{k} + 6 = 0 \Rightarrow 4\mathrm{k} = -4 \Rightarrow k = -1$.

4. $u = (3,\ 1)^T,\ v = (-2,\ 4)^T$ **uchun aniq** $\cos\theta ni$**, keyin** θ **ni gradus aniqligi bilan toping.**

[Manba: $ISALA$, hafta 2, mashq 4]

Yechim. $u \cdot v = 3 \cdot (-2) + 1 \cdot 4 = -6 + 4 = -2$; $\|u\| = \sqrt{10}, \|v\| = \sqrt{20} = 2\sqrt{5}$; $\cos\theta = -2/\sqrt{200} = -2/(10\sqrt{2}) = -\sqrt{2}/10 \approx -0.141$, demak $\theta \approx 98°$.

$$u \cdot v = 3 \cdot (-2) + 1 \cdot 4 = -6 + 4 = -2; ||u|| = \sqrt{10}, ||v|| = \sqrt{20} = 2\sqrt{5};$$
$$\cos\theta = (u \cdot v)/(||u|| \cdot ||v||) =$$

$$(-2)/(\sqrt{10} \cdot 2\sqrt{5}) = (-1)/(\sqrt{50}) = (-1)/(5\sqrt{2}) = -\sqrt{\frac{2}{10}}$$

$-\frac{1}{2} \approx -0.1414$, shuning uchun $\theta \approx 98^\circ$. $\sqrt{5}$

5. $u = (6,1)^T$ **vektorining** $v = (2,-1)^T$ **vektoriga proyeksiyasini toping;** $u - proj$ **ning** v **ga perpendikularligini tekshiring.**

[Manba: $ISALA$, hafta 2, mashq 5]

Yechim. $u \cdot v = 12 - 1 = 11, v \cdot v = 5$: $proj_v u = \frac{11}{5}(2,\ -1)^T = (22_5,\ -\frac{11}{5})^T$. Qoldiq: $u - proj = (\frac{8}{5}, 16_5)^T$; v bilan skalyar ko'paytma: $16_5 - \frac{16}{5} = 0$. Tekshirildi.

6. Nolga teng bo'lmagan p, q **vektorlari** $p \cdot q = \|p\|\|q\|$ **shartni qanoatlantiradi. Ular yo'nalishi haqida qanday xulosa chiqarish mumkin?**

[Manba: $ISALA$, hafta 2, mashq 6]

Yechim. $(45)dan, \cos\theta = 1$, demak $\theta = 0$: vektorlar bir xil yo'nalishda (biri ikkinchisining musbat karrali). Burchak formulasidagi tenglik Koshi-Bunykovskiy-Shvarts g'oyasining chegaraviy holati.

7. $a = (2,\ 1,\ -1)^T, b = (0,\ 3,\ 2)^T$ **uchun a×b va b×a ni hisoblang; belgilar munosabatini tekshiring.**

$$[2 \cdot 3 - (-1) \cdot 0; 5 - 5]$$

[Manba: $ISALA$, hafta 2, mashq 7]

Yechim. $a \times b = [1 \cdot 2 - (-1) \cdot 3;\ (-1) \cdot 0 - 2 \cdot 2;\ 2 \cdot 3 - 1 \cdot 0] = \begin{bmatrix} 5 \\ -4 \\ 6 \end{bmatrix}$, va $b \times a = \begin{bmatrix} -5 \\ 4 \\ -6 \end{bmatrix} =$ $-(a \times b)$ — ishora munosabatini tasdiqlaydi.

$$[2 \cdot 3 - 1 \cdot 0; 1 \cdot 0 - (-1) \cdot 3; (-1) \cdot 3 - 2 \cdot 2] = \begin{bmatrix} 6 \\ 3 \\ -7 \end{bmatrix}$$

$(48)ga$ mos keladi.

8. Mashq 7 dan olingan a **×b vektorining** $avab$ **ikkalasiga ham perpendikular ekanligini tekshiring.**

[Manba: $ISALA$, hafta 2, mashq 8]

Yechim. $a \cdot (a \times b) = 2 \cdot 5 + 1 \cdot (-4) + (-1) \cdot 6 = 10 - 4 - 6 = 0; b \cdot (a \times b) = 0 \cdot 5 + 3 \cdot (-4) + 2 \cdot 6 = -12 + 12 = 0$. Ikkalasi ham nolga teng, (47) dan keyin kafolatlanganidek.

9. $u = (1,\ 2,\ 2)^T, v = (3,\ -1,\ 0)^T$ **tomonli parallelogrammning yuzini toping.**

$$[2 \cdot 0 - 2 \cdot (-1); 2 \cdot 3 - 1 \cdot 0; 1 \cdot (-1) - 2 \cdot 3]$$

; yuz $= \sqrt{4} + 36 + 49 = \surd$

[Manba: $ISALA$, hafta 2, mashq 9]

Yechim. $u \times v = [2 \cdot 0 - 2 \cdot (-1);\ 2 \cdot 3 - 1 \cdot 0;\ 1 \cdot (-1) - 2 \cdot 3] = \begin{bmatrix} 2 \\ 6 \\ -7 \end{bmatrix}$, norma $= \sqrt{4} + 36 + 49 = \sqrt{89}$.

$$||u \times v|| = \sqrt{(2 \cdot 0 - 2 \cdot (-1))^2 + (2 \cdot 3 - 1 \cdot 0)^2 + (1 \cdot (-1) - 2 \cdot 3)^2} = \sqrt{4 + 36 + 49}$$

$$\sqrt{8}9$$

10. $P = (1,\ 0,\ 1)$, $Q = (3,\ 1,\ 0)$, $R = (0,\ 2,\ 4)$**:** PQR **uchburchakning yuzini toping.**

[Manba: *ISALA*, hafta 2, mashq 10]

Yechim. P dan chiqadigan tomonlar: $PQ = (2,\ 1,\ -1)^T$, $PR = (-1,\ 2,\ 3)^T$. Vektor ko'paytmasi: $PQ \times PR = (1 \cdot 3 - (-1) \cdot 2; (-1) \cdot (-1) - 2 \cdot 3; 2 \cdot 2 - 1 \cdot (-1)) = (5,\ -5,\ 5)$, norma

$$\sqrt{25 + 25 + 25} = \sqrt{75} = 5\sqrt{3}$$

; uchburchak yuzi $= (\frac{1}{2}) \cdot 5\sqrt{3} = 5\sqrt{3}/2$ (parallelogramm yuzining yarmi). $uvav$ ni almashtirish vektor ko'paytmasining ishorasini teskarisiga o'zgartiradi, yuz o'zgarmaydi.

11. $a = (1,\ t,\ 2)^T vab = (3,\ 6,\ 6)^T$ **vektorlari** a **×b** $= 0$ **shartni qanoatlantiradigan barcha haqiqiy** t **larni toping; agar bunday** t **lar bo'lmasa, tushuntiring.**

[Manba: *ISALA*, hafta 2, mashq 11]

Yechim. Parallel vektorlar nol vektor ko'paytmasini beradi, demak $a =$ cb bo'lishi kerak: $1 = 3\text{c} \Rightarrow c = \frac{1}{3}$; unda $t = 6\text{c} = 2$; uchinchisini tekshiramiz: $2 = 6\text{c} = 2$ — mos keladi. Shunday qilib $t = 2$ yagona qiymat (unda $a = \frac{1}{3}b$).

12. $a = (1,\ 0,\ 2)^T, b = (2,\ 1,\ -1)^T, c = (0,\ 3,\ 1)^T$**.** $a \cdot (b \times c)$ **ni hisoblang; nol qiymatini talqin qiling.**

$$1 \cdot 1 - (-1) \cdot 34$$

[Manba: *ISALA*, hafta 2, mashq 12]

Yechim. $b \times c = [1 \cdot 1 - (-1) \cdot 3;\ (-1) \cdot 0 - 2 \cdot 1;\ 2 \cdot 3 - 1 \cdot 0] = \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}$; $a \cdot (b \times c) = 1 \cdot 4 + 0 \cdot (-2) + 2 \cdot 6 =$ 16.

$$\begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & -1 \\ 0 & 3 & 1 \end{bmatrix}$$

Bu uch vektorga tortilgan parallelepipedning belgilangan hajmi; nol qiymati komplanarlikni (3D hajmi yo'qligini) anglatardi. Bu yerda $|16| = 16$ kub birlik. 2 −1 0

13. $A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 5 \end{bmatrix}$ **matritsasi berilgan. O'lchamini ko'rsating;** a_{12}, a_{23} **elementlarini toping; har bir qator va ustunni vektor sifatida yozing.**

[Manba: *ISALA*, hafta 2, mashq 13]

Yechim. O'lchami 2×3 (avval qatorlar). $a_{12} = -1 (1-$qator, 2-ustun$), a_{23} = 5$. Qatorlar: $\begin{bmatrix} 2 & -1 & 0 \end{bmatrix}, \begin{bmatrix} 4 & 3 & 5 \end{bmatrix}$; ustunlar: $\begin{bmatrix} 2 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \end{bmatrix}$.

$$[1][]35 - 1$$

14. $A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}$ **matritsalar berilgan.** $A + B, A - B, -$**2A, 3A − 2B ni hisoblang.**

[Manba: $ISALA$, hafta 2, mashq 14]

Yechim. $A+B=\begin{bmatrix}6 & 2\\ -2 & 6\end{bmatrix}$; $A-B=\begin{bmatrix}-4 & 4\\ -2 & 2\end{bmatrix}$; $-2A=\begin{bmatrix}-2 & -6\\ 4 & -8\end{bmatrix}$; $3A-2B=[3-10, 9+2;$ $-6-0,\ 12-4]=\begin{bmatrix}-7 & 11\\ -6 & 8\end{bmatrix}$.

15. A **matritsasi** 4×3 **o'lchamda,** B **—** $3\times 2, x\in R^2, y\in R^3, z\in R^4$**.** Ax, Ay, Az, Bx, By, Bz **dan qaysilari aniqlangan va ularning chiqish o'lchamlari qanday?**

[Manba: $ISALA$, hafta 2, mashq 15]

Yechim. Ay: $(4\times 3)(3\times 1)$ — aniqlangan, natija R^4. Bx: $(3\times 2)(2\times 1)$ — aniqlangan, natija R^3. Qolganlari (Ax, Az, By, Bz) ichki o'lchamlari mos kelmaydi va aniqlanmagan.

16. $D=diag(-2,\ 3,\ 1), x=(4,\ -1,\ 5)^T$ **berilgan.** Dx **ni hisoblang;** $Dhar$ **bir koordinataga nima qilishini tushuntiring.**

$$Dx=\begin{bmatrix}-2 & 0 & 0\\ 0 & 3 & 0\\ 0 & 0 & 1\end{bmatrix}, x=(2,-1,3)^T.$$

[Manba: $ISALA$, hafta 2, mashq 16]

Yechim. Dx: birinchi koordinata -2 ga masshtablanadi (aks etish bilan cho'zish), ikkinchi — $3ga$, uchinchi — 1/2 ga.

17. $A=$ **matritsasi berilgan.** $Axni(a)$ **ustunlarning 4 0 −1 kombinatsiyasi sifatida,** (b) **qatorlarning skalyar ko'paytmalari orqali hisoblang.**

$$1-22+2+9=13$$

[Manba: $ISALA$, hafta 2, mashq 17]

Yechim. $(a)2-1+3=$

$$2\begin{bmatrix}1\\2\\3\end{bmatrix}=\begin{bmatrix}2\\4\\6\end{bmatrix}-\begin{bmatrix}1\\2\\0\end{bmatrix}+\begin{bmatrix}3\\0\\3\end{bmatrix}$$
$$\begin{bmatrix}4\\2\\9\end{bmatrix}$$

. (b)1-qator: $1\cdot 2+(-2)(-1)+3\cdot 3=13$; 2-qator: $4\cdot 2+0\cdot(-1)+(-1)\cdot 3=5$; 3-qator: $2\cdot 2+2\cdot(-1)+2\cdot 3=6-2+6=10$.

$$\begin{bmatrix}1 & 4 & 2\\ -2 & 0 & 2\\ -1 & -1 & 2\end{bmatrix}\begin{bmatrix}2\\-1\\3\end{bmatrix}=\begin{bmatrix}13\\5\\10\end{bmatrix}$$

Xuddi shu natija, bo'lishi kerakgidek.

18. $a_1 = (1,\ 2)^T, a_2 = (3,\ -1)^T$ **vektorlar berilgan.** $x_1a_1 + x_2a_2 = (11,\ 1)^T$ **tenglamani qanoatlantiruvchi** $x = (x_1,\ x_2)^T$ **ni toping.**

[Manba: $ISALA$, hafta 2, mashq 18]

Yechim. Komponentlar: $x_1 + 3x_2 = 11$, $2x_1 - x_2 = 1$. Ikkinchisidan $x_2 = 2x_1 - 1$; keyin $x_1 + 6x_1 - 3 = 11 \Rightarrow 7x_1 = 14 \Rightarrow x_1 = 2$, $x_2 = 3$. 2 4

19. $A = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}$ **matritsasi berilgan.** $Ax = 0$ **tenglamani qanoatlantiruvchi barcha** x **ni tavsiflang; nima uchun ustunlar chiziqli bogliq?**

[Manba: $ISALA$, hafta 2, mashq 19]

Yechim. $2x_1 + 4x_2 = 0 va - x_1 - 2x_2 = 0$ — bir xil tenglama: $2x_1 = -4x_2$. Yechimlar: $x = t \cdot \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $t \in R$ — butun yechimlar to'g'ri chiziq. Trivial bo'lmagan kombinatsiya $-2 \cdot a_1 + 1 \cdot a_2 = 0$ mavjud, bu ustunlarning chiziqli bog'liqligidir (2-ustun 1-ustunning -2 barobariga teng).

$$[][]1 = (2, -1, 4)_T va A0 = (3, 2, 0)_T. Aniyozing$$

20. A **matritsasi** 3×2 **o'lchamda,** $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} vaA \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$**.** A **ni yozing va** $A \begin{bmatrix} -2 \\ 5 \end{bmatrix}$ **ni qator skalyar ko'paytmalarisiz hisoblang.**

$$A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \\ 5 & 4 \end{bmatrix}$$

[Manba: $ISALA$, hafta 2, mashq 20]

Yechim. Ikkita tasvir — bu A ning ustunlari: $A = -12$. Chiziqlilik: $A = 540$

$$\begin{bmatrix} 2 & 3 \\ 5 & 11 \end{bmatrix}$$

$-2Ae_1 + 5Ae_2 = -2 - 1 + 52 = 12$. 4 0 −8

21. Bir o'lchamli vektorlar uchun $(u + v) \cdot (u - v)$ **ni oching;** $u + v \perp u - v$ **bo'lishi uchun zarur va yetarli shart** $\|u\| = \|v\|$ **ekanligini isbotlang.**

[Manba: $ISALA$, hafta 2, mashq 21]

Yechim. Taqsimlab: $(u + v) \cdot (u - v) = u \cdot u - u \cdot v + v \cdot u - v \cdot v = \|u\|^2 - \|v\|^2$ (o'rtadagi hadlar bekor qilinadi). Perpendikulyarlik bu nolga teng bo'lishini anglatadi, ya'ni $\|u\|^2 = \|v\|^2$, va normalar manfiy bo'lmagani uchun $\|u\| = \|v\|$.

22. O'z-o'zini tekshirish: (1) skalyar ko'paytma va vektor ko'paytma qaysi ob'ektni beradi? $(2)u \cdot v$ **belgisi burchakni qanday tasniflaydi? (3) Nima uchun** 2×3 **matritsasi** R^3 **dan vektorni ko'paytiradi, lekin** R^2 **dan emas?** $(4)Axdax$ **komponentlari qanday ishlaydi? (5) 2-haftadagi qaysi amallar teng o'lchamli vektlarni talab qiladi, qaysilari aynan** R^3 **dan vektlarni?**

[Manba: $LAISA$, 2-hafta, o'z-o'zini tekshirish]

Yechim. (1) Skalyar; vektor (R^3da). (2) Musbat $\Rightarrow$ o'tkir, nol $\Rightarrow$ to'g'ri, manfiy $\rightarrow$ o'tmas. $(3)Ax$ ko'paytmasi ichki o'lchamlarning mosligini talab qiladi: $2\times 3va3\times 1$ ishlaydi; 2×1 esa yo'q. (4) Chiziqiy kombinatsiyadagi og'irliklar kabi $x_1a_1+\cdots+x_na_n$. (5) Skalyar ko'paytma, qo'shish va ayirish teng o'lchamli vektorlarni talab qiladi; vektor ko'paytmasi va uning geometriyasi (yuzalar, normalar) faqat R^3 da mavjud.

3.4 Chiziqli algebra I, 2-hafta: TA sessiyasi (15 masala) — savollar va to'liq yechimlar

$$[][]-25$$

1.$a=,b=$**: hisoblanga** $\cdot b$**; burchakni tasniflang. 4 1**

[Manba: $LATA,$2-hafta, masala 1]

Yechim. $a\cdot b=(-2)\cdot 5+4\cdot 1=-10+4=-6<0$: o'tmas (botiq).

2. $v=\begin{bmatrix}-2\\1\\2\end{bmatrix}$ **vektorining normasini va birlik vektorni toping.**

[Manba: $LATA,$2-hafta, masala 2]

Yechim. $\|v\|=\sqrt{4}+1+4=3$; birlik vektor $\hat{v}=[-\frac{2}{3}$; 1/3; 2/3].

3. $p=\begin{bmatrix}-2\\4\\3\end{bmatrix}$ $vaq=\begin{bmatrix}2\\1\\0\end{bmatrix}$ **orasidagi masofani toping.**

[Manba: $LATA,$2-hafta, masala 3]

Yechim. $p-q=\begin{bmatrix}-2 & 2\\4 & 1\\3 & 0\end{bmatrix}=\begin{bmatrix}-4\\3\\3\end{bmatrix}$, masofa $=\sqrt{16}+9+9=\sqrt{34}$.

4. Agar $\begin{bmatrix}2\\t\\4\end{bmatrix}$ **vektori** $\begin{bmatrix}1\\-3\\2\end{bmatrix}$ **ga perpendikulyar bo'lsa,** t **ni toping.**

[Manba: $LATA,$2-hafta, masala 4]

Yechim. $2-3\mathrm{t}+8=0\Rightarrow t=\frac{10}{3}$.

$$\begin{bmatrix}2\\-1\end{bmatrix}\cdot\begin{bmatrix}t\\-3\end{bmatrix}=0\Rightarrow 2t+3=0\Rightarrow$$
$$t=-\frac{3}{2}$$

5. $u=,v=$ **uchun** $\cos\theta$ **ni toping. 1 3**

$$\cos\theta=1/\sqrt{5}0=\sqrt{\frac{2}{10}}$$

[Manba: $LATA,$2-hafta, masala 5]

Yechim. $u \cdot v = 2 \cdot (-1) + 1 \cdot 3 = 1$; $\|u\| = \sqrt{5}, \|v\| = \sqrt{10}$; $\cos\theta = 1/\sqrt{50} = \sqrt{2}/10 \approx 0.141$, $\theta \approx 82°$.

6. $u = (5,\ -1)^T niv = (2,\ 1)^T$ **ga proyeksiya qiling;** u **ni parallel +** **perpendikulyar qismlarga yozing.**

$$[\frac{7}{5}; \frac{18}{5}]$$

[Manba: $LATA$, 2-hafta, masala 7]

Yechim. $u \cdot v = 9, v \cdot v = 5$: $proj =$

$$(\frac{9}{5}) \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

. Perpendikulyar qism:

$$\begin{bmatrix} 5 \\ -1 \end{bmatrix} - [\frac{14}{5}; \frac{9}{5}]$$

$u - proj =$

$$[\frac{11}{5}; -\frac{18}{5}]$$

. Yoyilma:

$$u = [\frac{14}{5}; \frac{9}{5}] + [\frac{11}{5}; -\frac{18}{5}]$$

.

7. $a = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$, $b = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$ **uchun** $a \times b$ **ni hisoblang.**

$$(-1) \cdot 1 - 3 \cdot 4 - 13$$

[Manba: $LATA$, 2-hafta, masala 8]

Yechim. $a \times b = [(-1) \cdot 1 - 3 \cdot 4;\ 3 \cdot 0 - 2 \cdot 1;\ 2 \cdot 4 - (-1) \cdot 0] = \begin{bmatrix} -13 \\ -2 \\ 8 \end{bmatrix}$.

$$|-48; 0 - 1|$$

8. $A = (0,\ 0,\ 0)$, $B = (2,\ 1,\ 0)$, $C = (1,\ 0,\ 3)$**: uchburchak yuzasini toping.**

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

[Manba: $LATA$, 2-hafta, masala 9]

Yechim. $AB = AC =$ vektor ko’paytmasi

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{1/2}$$

$= 0 \cdot 1 - 2 \cdot 3 = -6$, norma $= 46$; yuzasi $= \frac{46}{2}$.

$$S = (\frac{1}{2}) \cdot ||(2;1;0) \times (1;0;3)|| = (\frac{1}{2}) \cdot ||(3;-6;$$
$$-1)|| = (\frac{1}{2}) \cdot \sqrt{3^2 + (-6)^2 + (-1)^2} = (\frac{1}{2}) \cdot \sqrt{46}$$

9. $M =$ **uchun o'lcham,** m_{21}**, barcha ustunlar. 4 3 5**

$$\begin{bmatrix} 1 & 0 & -2 \\ 4 & 3 & 5 \end{bmatrix}$$

[Manba: $LATA$, 2-hafta, masala 10]

Yechim. 2×3; $m_{21} = 4$; ustunlar„. 4 3 5

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 3 \\ 5 & 0 & 0 \end{bmatrix}$$

10. $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ 2 & -2 \end{bmatrix}$**. 3A + 2B ni toping.**

[Manba: $LATA$, 2-hafta, masala 11]

Yechim. 3A $= \begin{bmatrix} 6 & 3 \\ -3 & 9 \end{bmatrix}$, 2B $= \begin{bmatrix} 0 & 8 \\ 4 & -4 \end{bmatrix}$; yig'indi $= \begin{bmatrix} 6 & 11 \\ 1 & 5 \end{bmatrix}$.

11. $A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{bmatrix}$, $x = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$**.** Ax **ni ustunlar kombinatsiyasi sifatida hisoblang.**

[Manba: $LATA$, 2-hafta, masala 12]

Yechim. $1 - 2 + 3 ==$.

$$\begin{bmatrix} 0 & 4 & 1 \\ -2 & 0 & 4 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 4 \\ 0 \\ 3 \end{bmatrix} + z \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} = [0x + 4y + 1z;$$
$$-2x + 0y + 4z;$$

$$1x + 3y + 2z] = \begin{bmatrix} 4y & + & z & \\ -2x & + & 4z & \\ x & + & 3y & + & 2z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 13 \end{bmatrix}$$

$$\begin{bmatrix}2 & -1\\ 3 & 2\end{bmatrix} + \begin{bmatrix}2\\ 2\end{bmatrix} + \begin{bmatrix}9\\ 13\end{bmatrix}$$

12. $a_1 =, a_2 =, b =$**.** $[a_1 a_2]x = b$ **bo'ladigan barcha** x **larni toping. 2 4 6**

[Manba: $LATA$, 2-hafta, masala 13]

Yechim. Ikkala tenglama ham $x_1 + 2x_2 = 3$ ga kelib chiqadi (tizma mos, lekin

$$\begin{bmatrix}3 & & 2t\\ -7 & 2 & \\ -9 & 4 & \end{bmatrix}$$

ortiqcha, chunki $a_2 = 2a_1 vab = 3a_1$). Umumiy yechim:$x =, t \in R$ — $tbir$ xil kombinatsiya uchun cheksiz ko'p og'irliklar.

13. Koordinatalarni ochmasdan, nima uchun $(u\times v)\cdot(u+v) = 0$ **ekanini tushuntiring.**

[Manba: $LATA$, 2-hafta, masala 14]

Yechim. Vektor ko'paytmasi ikkala kirish vektoriga ham perpendikulyar, shuning uchun ularning har qanday kombinatsiyasiga perpendikulyar: $(u\times v)\cdot u = 0 va (u\times v)\cdot v = 0$; qo'shib, $u+v$ bilan skalyar ko'paytma $0+0=0$ ga teng.

$$[u_1u_2u_3]\times[v_1v_2v_3]\cdot([u_1u_2u_3]+[v_1v_2v_3]) = 0$$

14. A **o'lchami** 2×2**, bunda** $A\cdot\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}2\\-1\end{bmatrix} vaA\cdot\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}3\\4\end{bmatrix}$**.** $AvaA\cdot\begin{bmatrix}5\\-2\end{bmatrix}$ **ni toping.**

[Manba: $TALA$, 2-hafta, masala 15]

Yechim. Standart bazis vektorlarining tasvirlari — ustunlar:

$$A = \begin{bmatrix}2 & 3\\ -1 & 4\end{bmatrix}$$

. U holda $A\cdot\begin{bmatrix}5\\-2\end{bmatrix} = 5\cdot\begin{bmatrix}2\\-1\end{bmatrix} - 2\cdot\begin{bmatrix}3\\4\end{bmatrix} = \begin{bmatrix}10 & 6\\ -5 & 8\end{bmatrix} = \begin{bmatrix}4\\-13\end{bmatrix}$; chiziqlilik hisobni skalyar ko'paytmasiz ikki masshtablashga aylantiradi.

4 3-hafta — Matritsa algebrasi, determinantlar, aralash ko'paytma (Chiziqli algebra I)

4.1 Chiziqli algebra I: 3-hafta formulalari katalogi

$$(A+B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c\,a_{ij}, \quad (A^T)_{ij} = a_{ji}, \quad (A^T)^T = A \tag{65}$$

$$(A+B)^T = A^T + B^T, \quad (cA)^T = cA^T, \quad A^T = A \text{ (simmetrik)} \tag{66}$$

$$A\in\mathbb{R}^{m\times n},\ B\in\mathbb{R}^{n\times p} \Rightarrow AB\in\mathbb{R}^{m\times p} \tag{67}$$

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} \tag{68}$$

$$A(BC) = (AB)C, \quad A(B+C) = AB + AC, \quad (A+B)C = AC + BC \tag{69}$$

$$(AB)^T = B^T A^T, \quad I_m A = A = A I_n \tag{70}$$

$$A\,[\,b_1 \cdots b_p\,] = [\,Ab_1 \cdots Ab_p\,] \text{ (ustunlar qoidasi)} \tag{71}$$

$$u^T v = \text{skalyar}, \quad uv^T = \text{matritsa (tashqi ko'paytma)} \tag{72}$$

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc \text{ (ishorali yuza)} \tag{73}$$

$$C_{ij} = (-1)^{i+j} \det M_{ij} \text{ (kofaktor)}, \quad \det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \tag{74}$$

bunda Mij — minori (i-qator va j-ustun o'chirilgan). Har qanday qator yoki ustun bo'ylab yoyish mumkin; nollari bor qator yoki ustunlar hisobni kamaytiradi. Uchburchak matritsalarda determinant diagonal elementlar ko'paytmasi. Boshlang'ich qator amallari qoidalari 1-jadvalda jamlangan.

A ustidagi amal	**det A ga ta'siri**
Ikkala qatorni almashtirish	ishorasini almashtiradi
Bitta qatorni c ga ko'paytirish	determinantni c ga ko'paytiradi
Bitta qatorning ko'paytmasini boshqasiga qo'shish	o'zgarish yo'q
Nol qator yoki ikki proporsional qator	determinant 0
Transponirlash	$\det(A^T) = \det A$
Bir o'lchamli kvadrat matritsalar ko'paytmasi	$\det(AB) = \det A \det B$

1-jadval. Kvadrat matritsalar uchun determinant qoidalari (qator versiyalari; ustun versiyalari ham o'rinli).

Butun $n \times n$ matritsani masshtablash n ta qatorning hammasini masshtablaydi: $\det(cA) = c^{\text{n}} \det A$. Determinantlar qo'shishga taqsimlanmaydi: $A = B = I_2$ uchun $\det(A+B) = 4$, holbuki $\det A + \det B = 2$.

$$(u \times v) \cdot w = u \cdot (v \times w) = \det[\,u\ v\ w\,] \text{ (A.K)} \tag{75}$$

$$V_{\text{parallelepiped}} = |(u \times v) \cdot w|, \quad h = \frac{|(u \times v) \cdot w|}{\|u \times v\|} \tag{76}$$

$$V_{\text{tetraedr}} = \frac{1}{6} |\det[\,u\ v\ w\,]|, \quad T = 0 \Leftrightarrow u, v, w \text{ komplanar} \tag{77}$$

4.2 Chiziqli algebra I: 3-hafta ma’ruzasi masalalari (15) — savollar va to’liq yechimlar

$$[][2]1-21$$

1. $A=\begin{bmatrix}1 & -2\\ 3 & 0\end{bmatrix}$, $B=\begin{bmatrix}2 & 1\\ -1 & 4\end{bmatrix}$ **bo’lsin. 2A** $-B$ **ni hisoblang va (2, 1) elementini ko’rsating;** $Ani2\times 3$ **matritsasiga qo’shish mumkinmi?**

[Manba: LA ma’ruzasi, 3-hafta, masala 1]

Yechim. 2A $=\begin{bmatrix}2 & -4\\ 6 & 0\end{bmatrix}$; 2A $-B=[2-2,-4-1;\ 6-(-1),\ 0-4]=\begin{bmatrix}0 & -5\\ 7 & -4\end{bmatrix}$; (2, 1) elementi $2(3)-(-1)=7$. 2×3 matritsasi bilan qo’shish aniqlanmagan — ikkala o’lcham ham mos kelishi kerak, umumiy elementlar soni emas.

2. $C=\begin{bmatrix}1 & 2 & 1\\ 0 & 3 & 2\end{bmatrix}$ **bo’lsin.** C^T **ni uning o’lchami bilan yozing;** CC^T **ni hisoblang;** u **simmetrikmi?**

[Manba: LA ma’ruzasi, 3-hafta, masala 2]

Yechim. $C^T=\begin{bmatrix}1 & 0\\ 2 & 3\\ 1 & 2\end{bmatrix}$ (3×2). $CC^T=[1+4+1,0+6-2;\ 0+6-2,\ 0+9+4]=\begin{bmatrix}6 & 4\\ 4 & 13\end{bmatrix}$. Diagonall elementlar — o’zlari bilan skalyar ko’paytirilgan qatorlar; tashqi diagonallari teng — matritsa simmetrik.

3. Yuqoridagi $Cvax=\begin{bmatrix}2\\ -1\\ 1\end{bmatrix}$ **uchun** Cx **ni ikki usul bilan hisoblang.**

[Manba: LA ma’ruzasi, 3-hafta, masala 3]

Yechim. Qatorlar: $1\cdot 2+2\cdot(-1)+(-1)\cdot 1=-1$; $0+3\cdot(-1)+2\cdot 1=-1$; demak $Cx=\begin{bmatrix}-1\\ -1\end{bmatrix}$. Ustunlar: $2\cdot\begin{bmatrix}1\\ 0\end{bmatrix}+(-1)\cdot\begin{bmatrix}2\\ 3\end{bmatrix}+1\cdot\begin{bmatrix}-1\\ 2\end{bmatrix}=\begin{bmatrix}-1\\ -1\end{bmatrix}$. O’lchamlar $(2\times 3)(3\times 1)2\times 1$ chiqish beradi.

Ustunlar: $2-+=$. O’lchamlar $(2\times 3)(3\times 1)$ chiqish 2×1 beradi.

$$C=\begin{bmatrix}0 & 3 & 2\\ -1 & 0 & 0\\ 0 & -1 & 0\end{bmatrix}$$

4. 1-masaladagi A,B **uchun** $ABvaBA$ **ni hisoblang. Tenglar?**

$$\begin{bmatrix}5 & 1 & 2\\ -2 & -1 & 1\\ 1 & -2 & 4\end{bmatrix}$$

[Manba: LA ma’ruzasi, 3-hafta, masala 4]

Yechim. $AB=$

$$\begin{bmatrix} 6 & 3 \\ 8 & 7 \end{bmatrix}$$

, $BA =$

$$\begin{bmatrix} 8 & 10 \\ 0 & 0 \end{bmatrix}$$

. Turlicha; hatto ikkala tartibda ham mavjud bo’lganda, tenglik kafolatlanmaydi.

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix},$$
$$D = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

5. $C = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 1 & -1 \\ 3 & 2 \end{bmatrix}$ **bo’lsin.** $CDvaDC$ **o’lchamlarini oldindan aytib, ikkalasini hisoblang.**

[Manba: LA ma’ruzasi, 3-hafta, masala 5]

Yechim. CD: $(2\times 3)(3\times 2) = 2\times 2$; DC: $(3\times 2)(2\times 3) = 3\times 3$. $CD = [2+2-3, 0-2-2;$ $0+3+6,\ 0-3+4] = \begin{bmatrix} 1 & -4 \\ 9 & 1 \end{bmatrix}$; $DC = \begin{bmatrix} 2 & 4 & -2 \\ 1 & -1 & -3 \\ 3 & 12 & 1 \end{bmatrix}$.

6. $S =$ **(siljish)**$, H =$ **(cho’zish)**$, x =$**. HSx va**

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & \end{bmatrix}$$

SHx ni hisoblang; har bir aylantirishni tavsiflang.

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

[Manba: LA darsi, 3-hafta, masala 6]

Yechim. $Sx =$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

, keyin HSx $=$

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

; shuningdek $Hx =$

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

, keyin SHx $=$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & \end{bmatrix}$$

. HS da siljish hissasi keyin cho'ziladi; $SHda$ esa avval cho'zish, keyin ustiga siljish qo'shiladi. Ikkala operatsiya bir xil, bosh tartibda, bosh natija: matritsalar almashmaydi.

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

,$v =$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

: hisoblangu$^T v$ vauv^T; tekshiring $(uv^T)^T = vu^T$.

7. $u = 20 - 13$ **uchun**

[Manba: LA darsi, 3-hafta, masala 7]

Yechim. Ichki: $u^T v = 2 + 0 - 3 = -1$ (skalyar). Tashqi: uv$^T =$

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 0 & 0 \\ 3 & 0 & -3 \end{bmatrix}$$

(3×3). Element bo'yicha transponirlab, $(uv^T)^T = (vu^T)_{ij}$, shuning uchun $(uv^T)^T = vu^T$. Yozilgan matritsalar — 3×3 o'zaro transponirlangan.

$$(AB)^T = B^T A^T$$

.

8. Masala $1danA, B$ **dan foydalanib,** $A^T B^T$ **mustaqil hisoblang; mos keladimi?**

$$[4]4 - 76$$

[Manba: LA darsi, 3-hafta, masala 8]

Yechim. $AB =$

$$\begin{bmatrix} 3 & 12 \\ -10 & -14 \end{bmatrix}$$

, demak $(AB)^T =$

$$\begin{bmatrix} 3 & -10 \\ 12 & -14 \end{bmatrix}$$

. Avval

$$\begin{bmatrix} 6 & 3 \\ -7 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 12 \\ -10 & -14 \end{bmatrix}$$

transponirlangan: $A^T =$

$$\begin{bmatrix} -2 & 0 \\ 1 & 4 \end{bmatrix}$$

, $B^T =$

$$\begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}$$

; keyin $A^T B^T =$

$$\begin{bmatrix} -2 & 0 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix} = \begin{bmatrix} -8 & -14 \\ 28 & -5 \end{bmatrix}$$

. Hisoblaymiz: $(2 \cdot 1 + (-1)(-2)) = 4$, $(2 \cdot 3 + 0) = 6$

$$\begin{bmatrix} 4 & 6 \\ 28 & -5 \end{bmatrix}$$

$= (AB)^T$, bu (70) ni tasdiqlaydi. Lekin $A^T B^T =$

$$\begin{bmatrix} -8 & -14 \\ 28 & -5 \end{bmatrix}$$

farq qiladi: to’g’ri identifikator ko’paytuvchilar tartibini o’zgartiradi.

9. $E = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ **deb belgilaymiz.** $\det E$ **ni hisoblang; uning ustunlariga tortilgan parallelogramm va uchburchkning maydonlarini toping; orientatsiya saqlanadimi?**

[Manba: LA darsi, 3-hafta, masala 9]

Yechim. $\det E = 3 \cdot 2 - 1 \cdot 1 = 5$. Parallelogramm maydoni $= |5| = 5$; uchburchk maydoni $= \frac{5}{2}$. Orientatsiya saqlanadi: $\det E > 0$.

10. $F = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ **uchun** $\det F$ **ni birinchi qator kofaktorlari bo'yicha hisoblang; birinchi ustun bo'yicha tekshiring.**

$$\begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix}$$

[Manba: LA darsi, 3-hafta, masala 10]

Yechim. Birinchi qator: $\det F = 2\det - 1\det + 0 = 2(6) - 1(-1) = 13.$ 0 2 1 2

$$\det F = 3\det \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} - 1\det \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} + 0\det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 3(0) - 1(0) + 0(1)$$

$$0.(3.1)$$

Birinchi ustun: $2\det + 1\det = 12 + 1 = 13$ (kofaktor (3, 1) belgisi $0231(-1)^{3+1} = +$). Har qanday qator yoki ustun bir xil qiymat beradi.

11. 3×3 **matritsa** M **uchun** $\det M = 5$ **ma'lum.** N **ni hosil qiling: 1,2-qatorlarni almashtiring; 3-qatorni 2 ga ko'paytiring;** $3\times$**(yangi 1-qator**$)ni2$**-qatorga qo'shing.** $\det N$, $\det(2M)$, $\det(M^T M)$ **ni toping.**

[Manba: LA darsi, 3-hafta, masala 11]

Yechim. Almashtirish: belgi o'zgaradi (-5); 3-qatorni masshtablash: yana ko'paytuvchi 2; qator qo'shish: o'zgarishsiz. $\det N = -2 \cdot 5 = -10$. Barcha qatorlarni masshtablash: $\det(2M) = 2^3 \cdot 5 = 40$. Ko'paytuvchilar qoidasi: $\det(M^T M) = \det M^T \det M = 25$.

$$\det(M^T M) = \det M^T \det M = 25.$$

$= 0, v = 2, w = 1$. Hisoblang $(u\times v) \cdot w$, hajmlar

12. $u103$ **paralleliped va tetraedr hajmlari; agaru,**v **ni almashtirsak nima o'zgaradi?**

$$[2 \cdot 2 - 0 \cdot 14]$$

[Manba: LA darsi, 3-hafta, masala 12]

Yechim. $u \times v = [1 \cdot 1 - 2 \cdot 0;\ 2 \cdot 0 - 1 \cdot 3;\ 1 \cdot 3 - 2 \cdot 0] = \begin{bmatrix} 1 \\ -3 \\ 3 \end{bmatrix}$; w bilan skalyar ko'paytma: $1 \cdot 2 + (-3) \cdot 1 + 3 \cdot 4 = 2$ - $3 + 12 = 11$. Hajm

$$2 \cdot 2 - 0 \cdot 14$$

paralleliped 11; tetraedr $\frac{11}{6}$. $uvav$ ni almashtirish belgini -11 ga o'zgartiradi, lekin ikkala hajm $11va\frac{11}{6}$ (absolyut qiymatlar) bo'lib qoladi.

$$0 \cdot 0 - 1 \cdot 2 = -2$$

.

13. $u = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $v = \begin{bmatrix}0\\2\\1\end{bmatrix}$, $w = \begin{bmatrix}1\\1\\k\end{bmatrix}$ **uchun vektorlar koplanar bo‘lgan barcha** k **ni toping; shu** $kdawniu, v$ **orqali ifodalang.**

$$0 \cdot 1 - 1 \cdot 2 - 2$$

$\Leftarrow\Rightarrow T = 0$**.** Bu **yerdau** $\times$**v** $= 1 \cdot 0 - 1 \cdot 1 = -1$**,**

[Manba: LA darsi, 3-hafta, masala 13]

Yechim. Komplanarlik

$$1 \cdot 2 - 0 \cdot 02$$

$T = -2 - 1 +$ 2k = 2k $-3 = 0 \Rightarrow k = \frac{3}{2}$ degani. U holdaw $= \alpha u + \beta v$: birinchi ikki koordinata $\alpha = 1, 2\beta = 1 \Rightarrow \beta = \frac{1}{2}$ beradi; uchinchisini tekshiramiz: $1 + 1_2 = 3_2$ — mos keladi,$w = u + \frac{1}{2}v$.

$$W = U + (\frac{5}{7})V,$$
$$\begin{bmatrix}2 & 0\\0 & 1\end{bmatrix} D = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix} (16)$$

14. $D = \begin{bmatrix}2 & 0\\0 & 3\end{bmatrix}$, $X = \begin{bmatrix}a & b\\c & d\end{bmatrix}$ **bo’lsin.** $DX = XD$ **bo’ladigan barcha** X **ni toping; agar** D **ning ikkala diagonal elementi teng bo’lsa nima bo’ladi?**

[Manba: LA darsi, 3-hafta, masala 14]

Yechim. $DX = \begin{bmatrix}2a & 2b\\3c & 3d\end{bmatrix}$, $XD = \begin{bmatrix}2a & 3b\\2c & 3d\end{bmatrix}$. Tenglik $b = c = 0$ talab qiladi, bunda $avad$ ixtiyoriy: aynan diagonal matritsalar bu D bilan kommutativ. Agar $D = \lambda I$ bo’lsa, u holda $DX = \lambda X = XD$ har qanday X uchun.

15. $u = \begin{bmatrix}2\\0\\1\end{bmatrix}$, $v = \begin{bmatrix}1\\2\\0\end{bmatrix}$, $w = \begin{bmatrix}0\\1\\3\end{bmatrix}$ $vaL = \begin{bmatrix}2 & 1 & 0\\0 & 1 & 0\\0 & 0 & -1\end{bmatrix}$ **bo’lsin. Qirralarni qayta hisoblamasdan,** $Lhar$ **bir qirraga qo’llanilgandan keyingi ishorali aralash ko’paytma va hajmni toping; hajm koeffitsienti nima uchun universal?**

$$\begin{bmatrix}1 & 0 & 3\\0 & 0 & -1\end{bmatrix}$$

belgili aralash ko’paytma va hajmdan $Lnihar$ **bir qirraga qo’llaganidan keyin toping; nima uchun hajm koeffitsienti universal?**

[Manba: LA darsi, 3-hafta, masala 15]

Yechim. $U = \begin{bmatrix} u & v & w \end{bmatrix}$ deb olaylik; 12-masaladan, $\det U = 13$. O'zgargan qirralar LU ni hosil qiladi, shuning uchun $\det(LU) = \det L \det U = (-2)(13) = -26$; yangi hajm 26. $|\det L| = 2$ koeffitsienti U dan mustaqil, chunki $\det(LU) = \det L \det U$ har qanday U uchun — determinantlar ko'paytmasi qoidasi geometrik ma'noni anglatadi: $|\det L|$ — hajmni o'zgartirish koeffitsienti.

5 4-hafta — Matritsa rangi va teskari matritsa (Lineal algebra I)

Lineal algebra I kursining 4-haftasi matritsa rangi va Gauss-Jordan usuli bilan matritsani teskarisini olishni o'z ichiga oladi, 1-3 haftalar bo'yicha Test 1 bilan tugaydi. Repozitoriyda 4-hafta uchun alohida mashqlar varaqasi yo'q, shuning uchun bu qism (i) Strangning berilgan materialidan to'plangan ikki mavzuga to'liq formulalar katalogi va $(ii)TA$ kursi varaqalari uslubida gradatsiyalashgan mashq to'plamini to'liq yechimlari bilan beradi. Ular kursning boshqa joylarida bir xil matematik tuzilmalardan foydalanadi, shuning uchun bir vaqtda 1-testga matritsa-algebra qismlari bo'yicha tayyorgarlik xizmatini qiladi.

5.1 4-hafta formulalar katalogi

6.1.1 Rang va yo'q qilish

$$\operatorname{rank} A \le \min(m, n) \tag{106}$$

$$A \in \mathbb{R}^{n\times n}:\ \operatorname{rank} A = n \Leftrightarrow Ax = 0 \text{ faqat } x = 0 \text{ yechimga ega} \tag{107}$$

$$A \text{ teskarililigi} \Leftrightarrow \operatorname{rank} A = n \Leftrightarrow n \text{ ta pivot} \Leftrightarrow \det A \neq 0 \tag{108}$$

$$A \text{ teskarililigi} \Leftrightarrow Ax = b \text{ har bir } b \text{ uchun aynan bitta yechimga ega} \tag{109}$$

6.1.2 Teskari matritsa (Strang, Introduction to Linear Algebra 4e, §2.5)

$$AA^{-1} = I = A^{-1}A; \quad x = A^{-1}b \text{ bo'lsa, } Ax = b \text{ yechim} \tag{110}$$

$$A^{-1} \text{ mavjud} \Leftrightarrow \text{yo'q qilish } n \text{ ta pivot beradi (qator almashtirishlar mumkin)} \tag{111}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (ad - bc \neq 0) \tag{112}$$

$$(AB)^{-1} = B^{-1}A^{-1}; \quad (ABC)^{-1} = C^{-1}B^{-1}A^{-1} \text{ (teskari tartib)} \tag{113}$$

$$\operatorname{diag}(d_1, \ldots, d_n)^{-1} = \operatorname{diag}\left(\frac{1}{d_1}, \ldots, \frac{1}{d_n}\right) \ (d_i \neq 0) \tag{114}$$

$$x \neq 0,\ Ax = 0 \Rightarrow A \text{ teskarili emas} \tag{115}$$

$$BA = I,\ AC = I \Rightarrow B = B(AC) = (BA)C = C \text{ (yagonalik)} \tag{116}$$

(115) ning izohi: hech bir matritsa 0 ni orqaga x ga qaytara olmaydi.

6.1.3 Gauss–Jordan yo'q qilish

$$[\,A \mid I\,] \xrightarrow{\text{boshlang'ich qator amallari}} [\,I \mid A^{-1}\,] \tag{117}$$

Bir xil qator amallari $AniIgavaIniA^{-1}$ ga aylantiradi. Qator amallari: ikkala qatorni almashtirish; qatorni noldan farqli konstantaga ko'paytirish; bitta qatorning ko'paytmasini boshqasiga qo'shish. Har biri teskari oson (amalni bekor qilish) bo'lgan boshlang'ich E matritsaga mos keladi, shuning uchun $(E_2E_1A)^{-1} = A^{-1}E_1^{-1}E_2^{-1}$ teskari tartibda chiqadi.

5.2 4-hafta mashqlari (rang, teskari, Gauss-Jordan) — savollar va to'liq yechimlar

1. $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 5 \\ 3 & 6 & 4 \end{bmatrix}$ **uchun chiqarib tashlash orqali** $\operatorname{rank} A$ **ni hisoblang.**

[Manba: LA4-hafta amaliyoti, masala 1]

Yechim. Birinchi ustun elementi (1) ostida chiqaramiz: $R_2 - 2R_1 \to (0, 0, 3)$; $R_3 - 3R_1 \to (0, 0, 1)$. Ikkinchi ustun elementlari yo'qoldi, demak ikkinchi ustun elementi 3-ustundan keladi: ustun elementi $= 3$. Uning ostida chiqaramiz: $R_3 - \frac{1}{3}R_2 \to (0, 0, 0)$. Ikkita nol bo'lmagan ustun element: $\operatorname{rank} A = 2$. E'tibor bering $col_2 = 2col_1$, bu boshidan ko'rinib turardi — rang kamchiligi belgisi. 2 1

2. $B =$**, uchta usul bilan teskari ekanligini aniqlang: ustun elementlar, 4 2 determinant va nol bo'lmani nol vektor.**

[Manba: LAming 4-haftalik amaliyoti, 2-masala]

Yechim. (Asosiy)$R_2 - 2R_1 \to (0, 0)$: faqat bitta asosiy — noyob. (Determinant)$ad - bc = 2 \cdot 2 - 1 \cdot 4 = 0$ — (112) bo'yicha noyob. (Nol-vektor)$B(1, -2)^T = (2-2, 4-4)^T = 0$ — $Bx = 0$ tenglamaning nol bo'lmagan yechimi, demak (115) bo'yicha B teskarisiz. Uchala test ham mos keladi. 212×2 formulasi bo'yicha hisoblang va $AA^{-1} = I$ ekanligini tekshiring.

3. $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ **matritsasining teskarisini toping.**

[Manba: LAming 4-haftalik amaliyoti, 3-masala]

Yechim. $\det A = 6 - 5 = 1$, shuning uchun (112) bo'yicha $A - 1 = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$. Tekshirish: $A \cdot A - 1 = [2 \cdot 3 + 1 \cdot (-5),\ 2 \cdot (-1) + 1 \cdot 2;\ 5 \cdot 3 + 3 \cdot (-5),\ 5 \cdot (-1) + 3 \cdot 2] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$.

$$A^{-1} = (\frac{1}{7}) \cdot \begin{bmatrix} 2 & -3 \\ 1 & 5 \end{bmatrix}$$

$$A = [5 \cdot 3 + 3 \cdot (-5) 5 \cdot (-1) + 3 \cdot 2; 01] = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

4. $\begin{bmatrix} A & I \end{bmatrix}$ **matritsiga Gaus-Jordan usulini qo‘llab** $A =$**. 3 7 matritsasini teskarisini toping.**

[Manba: LAming 4-haftalik amaliyoti, 4-masala]

Yechim.

$$\begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} \to R_2 - 3R_1 \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \to R_1 - 2R_2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$A^{-1} = \begin{bmatrix} -7 & 2 \\ 3 & -1 \end{bmatrix}$$

Shunday qilib $A^{-1} =$; tekshirish: $AA^{-1} ==$.

1

5. Gaus-Jordan usuli bilan teskarisini toping va tekshiring, 0 −1 2 asosiy ko‘paytmalari $\det K$ **ga teng ekanligini.**

[Manba: LAming 4-haftalik amaliyoti, 5-masala (Strangning sevimli misoli)]

Yechim. $[K|I]$ da ishlaymiz: birinchi opor 2 bilan chiqaramiz $(R2 + (\frac{1}{2})R1)$, so’ngra ikkinchi opor $\frac{3}{2}(R3 + (\frac{2}{3})R2)$, so’ngra uchinchi 4/3; yuqoriga davom etib, oporlar ustidini tozalaganda o’ngda $K - 1$ hosil bo’ladi.

$$K^{-1} = (\frac{1}{4}) \cdot \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

bo‘ladi. Asosiy ko‘paytmalar $2 \cdot \frac{3}{2} \cdot 4_3 = 4 = \det K$ ga teng, va haqiqatan ham K^{-1} butun sonli matritsani $\det K$ ga bo‘ladi — (112) dan keyin belgilangan namuna. (K simmetrik va uch diagonal, lekin K^{-1} zich.)

4-masaladagi A^{-1} yordamida.

6. $Ax = b$ **tenglamani** $A =, b = 371$

uchun yeching.

[Manba: LAming 4-haftalik amaliyoti, 6-masala]

Yechim. $x = A{-}1\text{b} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = [7 \cdot 1 - 2 \cdot 1;\ -3 \cdot 1 + 1 \cdot 1] = \begin{bmatrix} 5 \\ -2 \end{bmatrix}.$

$$A^{-1} = (1)/(7) \begin{bmatrix} 1 & -2 \\ 3 & 1 \end{bmatrix}$$

; $(AB)^{-1}$ ni avval

7. $A = vaB = 0211$ **matritsalar berilgan.** $(AB)^{-1}$ **ni avval**

[Manba: LAming 4-haftalik amaliyoti, 7-masala]

Yechim. (113) formulaga ko'ra, $(AB)^{-1} = B^{-1}A^{-1}$. Bu yerda $\det A = 2va \det B = 2$, shuning uchun

$$det(AB) = 4, teskarimatritsani(AB)^{-1} = (1)/(4)\begin{bmatrix} 2 & -3 \\ -2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}[(-121)/2(21)/(-2);$$

$$(-314)/4(21)/(-2)] = [\frac{1}{2}\frac{1}{2};$$

$$\frac{3}{4}\frac{1}{4}]—birxilmatritsa.Va$$

$(AB) - 1(AB) = [\frac{1}{2}\frac{1}{2};\ \frac{3}{4}\frac{1}{4}]\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = [\frac{1}{2}\frac{1}{2};$ 3/4 1/4] * [(−1 2 1)/2 (2 1)/(−2); $(-314)/4(21)/(-2)] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Teskari tartib qonuni tasdiqlandi.

8. (Tushunchaviy, Test-1 uslubida.) To'g'ri yoki noto'g'ri, asoslari bilan: (a) **Agar** A **matritsa teskari bo'lsa,** A^T **ham teskari bo'ladi.** (b) **Agar** $Ax = \lambda xx \neq 0va\lambda = 0$ **bo'lsa,** A **teskari emas.** (c) **Teskari** 3×3 **matritsalar ko'paytmasi noyob bo'lishi mumkin.** (d) **Har qanday** 2×3 **matritsa uchun** $\text{rank}(A) \leq 2$**.**

[Manba: LAming 4-haftalik amaliyoti, 8-masala]

Yechim. (a) To'g'ri: $Adan$ ta asosli element bo'lsa $\Longleftrightarrow$ A^Tdan ta asosli element bo'ladi (qator rangi = ustun rangi); rasmiy ravishda $(A^T)^{-1} = (A^{-1})^T$. (b) To'g'ri: $Ax = 0 \cdot x = 0$ tenglamani qanoatlantiruvchi nol bo'lmagan $x(115)$ shartni buzadi. (c) Noto'g'ri: (113) formulaga ko'ra teskari matritsalar ko'paytmasi ham teskari bo'ladi. (d) To'g'ri: rang matritsa satrlari sonidan oshmaydi (har bir asosli element bitta satrni egallaydi), shuning uchun rank $\leq 2 < 3$.

9. (Amaliy, Test-1 uslubida.) c **ning qanday qiymat**$(lar)ida A = \begin{bmatrix} 1 & c \\ 4 & 2 \end{bmatrix}$ **matritsasi yagona (singulyar emas)? Qolgan qiymatlar uchun** $Ax = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$ **ni yeching.**

[Manba: LAming 4-haftalik amaliyoti, 9-masala]

Yechim. $\det A = c - 6$; noyob bo'lishi uchun aynan $c = 6$. Aks holda $A^{-1} =^1$

$$\det A = (c - 6)/(4c - 20)$$

vax $= A^{-1}b =^1$, masalan $c = 7$ da:$x = (8, -2)^T$.

$$x = A^{-1}b = (beginarrayc8 - 2endarray).$$

10. (Yuqori murakkablik.) Isbotlang, agar A **teskari bo'lsa,** u **holda** $\text{rank}(AB) = \text{rank}(B)$ **har qanday mos keladigan** B **uchun.**

[Manba: LAm 4-hafta amaliyoti, 10-masala]

Yechim. Teskari matritsaga ko'paytirish soni asosli vektorlarni o'zgartirmaydi: standavy fikrlash: rank$(AB) \leq$ rank$B(AB$ qatorlari — B qatorlarining chiziqli kombinatsiyalari), va xuddi shuni $B = A^{-1}(AB)$ uchun qo'llaymiz, rank$B \leq$ rank(AB) hosil bo'ladi; ikkala tengsizlik tenglikni talab qiladi. Intuitsiya: teskari ko'paytuvchi faqat fazani cho'zadi/aylantiradi/siljitadi, lekin tekislamaydi, shuning uchun rang yo'qolishi yoki paydo bo'lishi mumkin emas.

6 Darsliklardan qo'shimchalar — to'liq yechimlari bilan majburiy o'qish uchun qo'shimcha mashqlar

6.1 Strang, Introduction to Linear Algebra 4e — tanlangan masalalar yechilgan

1. (§1.1) Barcha cu ni tavsiflang $u = (2, 1)$ uchun va barcha cu + dv kombinatsiyalarini $u = (2,\ 1)$, $v = (1,\ 2)$ uchun; qachon cu + dv kombinatsiyalari tekislikni to'ldiradi?

[Manba: Strang Intro 4e, g'oyalar to'plami §1.1]

Yechim. cu (2, 1) orqali o'tgan to'g'ri chiziqni to'ldiradi: undagi har bir vektor — uning biror karrali. cu + dv koordinata boshidan o'tgan tekislikni to'ldiradi, agar $21u$ vav kollinear bo'lmasa; bu yerda det $= 3 \neq 0$, shuning uchun ikki yo'nalat 1 2 mustaqil va kombinatsiyalar butun R^2 ni to'ldiradi. Aynan shu mezon — nol bo'lmaydigan determinant — 4-haftada teskarilik sifatida qaytadi (108).

2. (§1.2) Skalyar ko'paytmalarni hisoblang:$u \cdot v$ uchunu $= (1,\ 2,\ 1), v = (3,\ 0,\ 2)$; ikkalasiga perpendikulyarw vektorni toping.

[Manba: Strang Intro 4e, masalar to'plami §1.2]

Yechim. $u \cdot v = 3 + 0 + 2 = 5$. Perpendikulyar vektor — vektor ko'paytmasi:$w = u \times$v $=$

$$\begin{bmatrix}1\\2\\1\end{bmatrix} \times \begin{bmatrix}3\\0\\2\end{bmatrix} = [(2 \cdot 2 - 1 \cdot 0);$$

$$(1 \cdot 3 - 1 \cdot 2); (1 \cdot 0 - 2 \cdot 3)] = \begin{bmatrix}4\\1\\-6\end{bmatrix}$$

; tekshirish: $w \cdot u = 4 + 2 - 6 = 0 va w \cdot v = 12 + 0 - 12 = 0$.

3. (§1.2) Qachonu $\cdot v = v \cdot u$ manfiy va bu burchak haqida nima deydi? Nolga teng qachon?

[Manba: Strang Intro 4e, tushuncha §1.2]

Yechim. Manfiy aynan $\cos\theta < 0$ bo'lganda, ya'ni burchak 90° dan katta (o'tkir); nol $\theta = 90^\circ$ da — perpendikulyarlik. Bu tasnif, 2-haftadagi TA masalarida qo'llanilgan. $21, b = (3, 4)^T$ orqali A^{-1}.

4. $(§2.5) Ax = b$ **ni yeching** $A = 12$

$$[2-12;3-42;\frac{2}{3}]$$

[Manba: Strang Intro 4e, masalar to'plami §2.5]

Yechim. $\det A = 3$: $A^{-1} = {}^{1}; x = A^{-1}b = {}^{1}={}^{1}=.$

$$detA = 3; A = [1-12;3-38;35\frac{5}{3}]; x = A^{-1}b = [\frac{1}{4};4;$$
$$2] = [\frac{14}{4};4;2] = [\frac{7}{2};4;2](16)$$

Tekshirish: $2\cdot(\frac{2}{3})+5\cdot(\frac{1}{3})=3$, $2\cdot(\frac{1}{3})+2\cdot(\frac{5}{3})=4$.

5. (§2.5) Nol bo'lmagan $Ax=0$ yechimi teskarini istisno qiladi? Ikkita jumladan iborat fikrni keltiring, konsept «hech qanday matritsa$0nix$ ga qaytara olmaydi» deb nomlanadi.

[Manba: Strang Intro 4e, §$2.5ga4$-e'lon]

Yechim. Agar $Ax = 0x \neq 0vaA$ da teskari bo'lsa, u holdax $= A^{-1}Ax = A^{-1}0 = 0$ — ziddiyat. Nol vektorning «teskari tasviri» faqat aynilikda mavjud; nol-vektor buni emasligini isbotlaydi.

6. (§2.5) Agar $AvaC$ teskari 3×3 matritsalar bo'lsa, $A+C$ teskari bo'ladimi? AC? $A^{-1}C$?

[Manba: Strang Intro 4e, masalar to'plami §2.5]

Yechim. $A+C$: albatta emas (masalan, $C=-A0$ beradi); teskari matritsalarning yig'indisi aynan bo'lishi mumkin. AC: ha, $(AC)^{-1}=C^{-1}A^{-1}(113)$. $A^{-1}C$: ha, ikkita teskari matritsaning ko'paytmasi.

6.2 Strang, Linear Algebra and its Applications 4e, §1.6 (Teskari va transponirlangan) — asosiy natijalar va tahlil qilingan misol

Belgilangan o'qish §1.6 katalogga uchta qo'shimcha element qo'shadi: Gauss– 2 1 1

$$\det A^{-1} = (1)/(\det A)$$

Jordani bilan $A = 4-60_{detA}$ va transponirlanish -2 7 2 identifikatorlari. 2 1 1

1. $A=\begin{bmatrix}2&1&1\\4&-6&0\\-2&7&2\end{bmatrix}$ uchun Gauss-Jordan usulini bajaring va $A-1$ ni ko'rsating.

[Manba: Strang LAiA 4e, §1.6, misol 1]

Yechim. I ni to'ldirib, chiqarib tashlaymiz: asosiy $2,-8,1(R_2-2R_1,\ R_3+R_1$, keyin $R_3+\frac{3}{4}R_2$ amallaridan so'ng). Yuqoriga davom etib, asosiy elementlar ustini tozalab, satrlarni 2, -8, 1 ga bo'lib,

$$A^{-1}=(\frac{1}{8})\begin{bmatrix}12&-5&-6\\8&-3&-2\\-1&1&1\end{bmatrix}$$

hosil qilamiz. Determinant asosiy elementlar ko'paytmasiga teng $2(-8)(1) = -16$, va haqiqatan ham $A^{-1}\det A$ ga bo'lishni ko'rsatadi (kofaktorlarning belgilari hisobga olingan — $A^{-1} = \frac{adjA}{detA}$ ga ekvivalent).

2. §1.6 da qo'llanilgan determinant-transponirliqlik identifikatsiyalarini ko'rsating: $\det A^T$, $\det(AB)$, $\det A^{-1}$**.**

[Manba: Strang LAiA 4e, §1.6]

Yechim. $\det A^T = \det A$; $\det(AB) = \det A \det B$(1–jadval); va $B = A^{-1}$ deb olib, $\det(A)\det(A^{-1}) = \det I = 1$, shuning uchun $1 \det A^{-1} = \det A$, bu esa "teskari mavjud $\Leftrightarrow\Rightarrow$ $\det A \neq 0$" degani aniq, determinantlar katalogini (3-hafta) teskari matritsalar katalogi (4-hafta) bilan bog'laydi.

3. Isbotlang $(A^{-1})^T(A^T)^{-1} va A^T A$ **simmetrik ekanini tushuntiring, manfiy bo'lmagan diagonalga ega.**

[Manba: Strang LAiA 4e, §1.6]

Yechim. $AA^{-1} = I$ ni transponirib, $(A^{-1})^T A^T = I$ hosil bo'ladi, bu esa

$$(A^{-1})^T = (A^T)^{-1}(16)$$

degani aniq: Simmetriya: $= A^T A$; diagonal elementlari P

$$(A^T)^{-1} = (A^{-1})^T; matritsa A^T A simmetrik,$$
$$(A^T A)^T = A^T A, uning diagonal elementlari manfiy$$

$$bo'lmagan : a_{ii} = \Sigma \frac{n}{k=1} a_{ki}^2 \geq 0 — kvadratlar yig'indisi — fakt.(17)$$

3-hafta ma'ruzalaridagi CC^T bilan bog'liq.